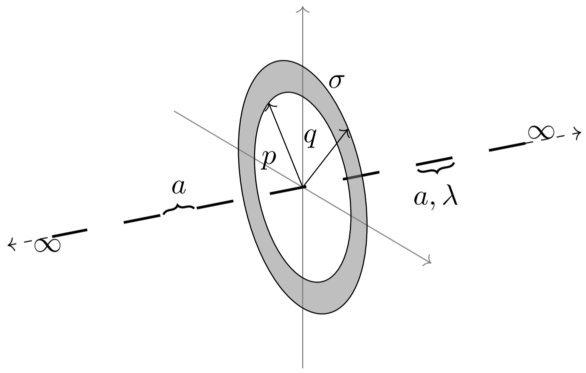
1. Find the net force on ring due to the following line charge distribution. The ring has a uniform charge density $\sigma \ \frac{C}{m^2}$, with inner radius p and outer radius q. The length of each line segment is a (in meter), with linear charge distribution $\lambda \ \frac{C}{m}$, the distance between each line segment is also a. $\left(\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{\sqrt{a^2+(nb)^2}} = \frac{\pi}{b} \frac{1}{\sinh\left(\frac{\pi a}{b}\right)}\right)$



Solution:

To solve this we need to know the field acted on a point charge due a general line segment of length 'a' and linear charge distribution 'λ'. The system will be something like this

1. Electric Field from a Single Side Consider the semi-infinite array of segments on the positive z-axis. The segments are located at intervals $[a, 2a], [3a, 4a], \ldots$ with gaps of length a in between. The electric field on the axis of a charged line segment from z_1 to z_2 at a transverse distance r is:

$$E_z(r) = \frac{k\lambda}{r}\left[\frac{z}{\sqrt{r^2+z^2}}\right]_{z_1}^{z_2} \times \frac{r}{z} \ \text{(axial component)} \implies E_z = k\lambda\left(\frac{1}{\sqrt{r^2+z_1^2}} - \frac{1}{\sqrt{r^2+z_2^2}}\right)$$

Summing over all segments $(z_1 = (2m-1)a, z_2 = 2ma)$:

$$E_{\text{right}} = k\lambda \sum_{m=1}^{\infty}\left(\frac{1}{\sqrt{r^2+((2m-1)a)^2}} - \frac{1}{\sqrt{r^2+(2ma)^2}}\right)$$

This series can be rewritten as an alternating sum:

$$E_{\text{right}} = -k\lambda \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{r^2+(na)^2}}$$

2. Applying the Summation Identity

$$S_{\text{total}} = \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{\sqrt{r^2+(na)^2}} = \frac{\pi}{a}\frac{1}{\sinh\left(\frac{\pi r}{a}\right)}$$

We can relate our one-sided sum to this total sum. Note that $S_{\text{total}} = \text{Term}_0 + 2\sum_{n=1}^{\infty} \ldots$, where $\text{Term}_0 = 1/r$.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{r^2 + (na)^2}} = \frac{1}{2}\left(S_{\text{total}} - \frac{1}{r}\right) = \frac{1}{2}\left(\frac{\pi}{a\sinh\left(\frac{\pi r}{a}\right)} - \frac{1}{r}\right)$$

Substituting this back into the field expression:

$$|E_{\text{right}}| = \frac{k\lambda}{2}\left(\frac{1}{r} - \frac{\pi}{a\sinh\left(\frac{\pi r}{a}\right)}\right)$$

3. Calculation of Net Force The force on the ring is found by integrating the field over the charge distribution of the ring ($dq = \sigma 2\pi r dr$). Assuming contributions from both sides add constructively:

$$F_{\text{net}} = 2\int_p^q |E_{\text{right}}|(2\pi r\sigma)dr = 2\pi\sigma\lambda k\int_p^q 2r \cdot \frac{1}{2}\left(\frac{1}{r} - \frac{\pi}{a\sinh\left(\frac{\pi r}{a}\right)}\right)dr$$

Simplifying the integrand:

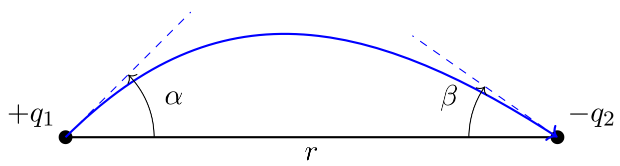
$$F_{\text{net}} = \frac{\sigma\lambda}{2\epsilon_0}\int_p^q \left(1 - \frac{\pi r}{a\sinh\left(\frac{\pi r}{a}\right)}\right)dr$$

Evaluating the first term and substituting $u = \frac{\pi r}{a}$ for the second:

$$\boxed{F_{\text{net}} = \frac{\sigma\lambda}{2\epsilon_0}\left[(q-p) - \frac{a}{\pi}\int_{\frac{\pi p}{a}}^{\frac{\pi q}{a}} \frac{u}{\sinh u}\,du\right]}$$

Note: The integral $\int \frac{u}{\sinh u}du$ results in non-elementary functions (related to Polylogarithms), so the solution is typically left in this definite integral form.

2. Consider 2 point charges of magnitude q_1 and $-q_2$ placed at a distance r from each other suppose that an electric field line emerges from q_1 at an angle α with the line joining the 2 charges and terminates into $-q_2$ at an angle β. Find the relation governing the 2 angles.



Solution:

The electric flux Φ emanating from a point charge q into a solid angle Ω is given by Gauss's Law as:

$$\Phi = \frac{q}{\varepsilon_0}\frac{\Omega}{4\pi}$$

The solid angle Ω of a cone with semi-vertical angle θ is $\Omega = 2\pi(1-\cos\theta)$. Therefore, the flux contained within a cone of angle θ from a point charge q is:

$$\Phi(\theta) = \frac{q}{2\varepsilon_0}(1-\cos\theta) = \frac{q}{\varepsilon_0}\sin^2\left(\frac{\theta}{2}\right)$$

Since electric field lines do not cross, the flux remains conserved within the "tube" of field lines bounded by the rotation of the given field line around the axis joining the charges.

The flux leaving $+q_1$ within the cone of semi-vertical angle α must equal the flux entering $-q_2$ within the cone of semi-vertical angle β. Thus:

$$\Phi_{out}(q_1) = \Phi_{in}(q_2)$$

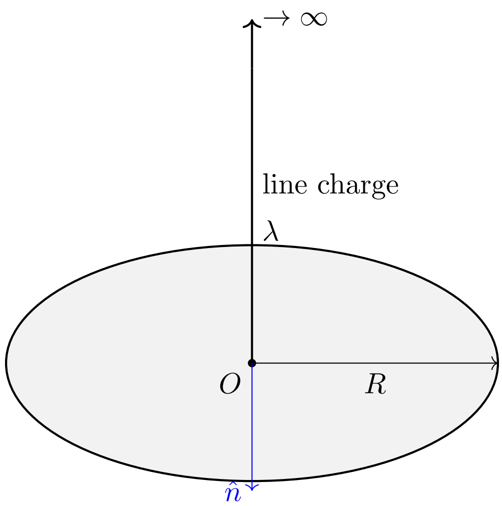
$$\frac{q_1}{2\varepsilon_0}(1-\cos\alpha) = \frac{q_2}{2\varepsilon_0}(1-\cos\beta)$$

Using the half-angle identity $1-\cos\theta = 2\sin^2(\frac{\theta}{2})$, we get:

$$q_1 \cdot 2\sin^2\left(\frac{\alpha}{2}\right) = q_2 \cdot 2\sin^2\left(\frac{\beta}{2}\right)$$

$$\boxed{q_1\sin^2\left(\frac{\alpha}{2}\right) = q_2\sin^2\left(\frac{\beta}{2}\right)}$$

3. Consider a semi-infinite line charge with linear charge density $\lambda\ C/m$ which rests on a circular disc of radius R (a circular region of free space). Compute the flux passing through the circular disc.



Solution:

We calculate the total flux by integrating the contributions of infinitesimal charge elements along the line charge.

Consider a small element of the line charge of length dz at a height z from the center of the disc $(0 \le z < \infty)$. The charge of this element is $dq = \lambda\, dz$.

The electric flux $d\Phi$ through a disc of radius R due to a point charge dq located at a distance z on its axis is given by the solid angle formula (derived from the solid angle $\Omega = 2\pi(1-\cos\theta)$ subtended by the disc at z):

$$d\Phi = \frac{dq}{2\varepsilon_0}(1-\cos\theta)$$

where θ is the semi-vertical angle of the cone formed by the disc and the point charge. From geometry, $\cos\theta = \frac{z}{\sqrt{R^2+z^2}}$. Substituting this into the flux expression:

$$d\Phi = \frac{\lambda\, dz}{2\varepsilon_0}\left(1 - \frac{z}{\sqrt{R^2+z^2}}\right)$$

To find the total flux Φ, we integrate from $z=0$ to $z \to \infty$:

$$\Phi = \int_0^\infty \frac{\lambda}{2\varepsilon_0}\left(1 - \frac{z}{\sqrt{R^2+z^2}}\right) dz$$

$$\Phi = \frac{\lambda}{2\varepsilon_0}\left[z - \sqrt{R^2+z^2}\right]_0^\infty$$

Evaluating the limits:

- At the lower limit $(z=0)$: Value is $0 - \sqrt{R^2} = -R$.
- At the upper limit $(z \to \infty)$: We examine the behavior of $z - \sqrt{z^2+R^2}$:

$$\lim_{z\to\infty}\left(z - z\sqrt{1+\frac{R^2}{z^2}}\right) \approx \lim_{z\to\infty}\left(z - z\left(1+\frac{R^2}{2z^2}\right)\right) = \lim_{z\to\infty}\left(-\frac{R^2}{2z}\right) = 0$$

Thus, the definite integral value is:

$$\left[z - \sqrt{R^2+z^2}\right]_0^\infty = 0 - (-R) = R$$

Substituting this back into the expression for Φ:

$$\Phi = \frac{\lambda}{2\varepsilon_0}\cdot R$$

$$\boxed{\Phi = \frac{\lambda R}{2\varepsilon_0}\frac{Nm^2}{C}}$$

4. Consider the following potential distribution in spherical coordinates

$$\Phi(r,\theta) = \frac{V_0}{a}\left(r - \frac{r^2}{2a} + \frac{a(3\cos^2\theta - 1)}{6}\right)$$

What is the electric field and where is its maximum

Solution:

The electric field is

$$\vec{E} = -\nabla\Phi = -\frac{\partial\Phi}{\partial r}\hat{r} - \frac{1}{r}\frac{\partial\Phi}{\partial\theta}\hat{\theta}$$

Compute derivatives:

$$\frac{\partial\Phi}{\partial r} = \frac{V_0}{a}\left(1 - \frac{r}{a}\right), \quad \frac{\partial\Phi}{\partial\theta} = \frac{V_0}{6}\cdot 6\cos\theta(-\sin\theta) = -V_0\cos\theta\sin\theta$$

Hence,

$$\boxed{\vec{E}(r,\theta) = \frac{V_0}{a}\left(\frac{r}{a} - 1\right)\hat{r} + \frac{V_0}{r}\cos\theta\sin\theta\,\hat{\theta}}$$

Magnitude of the electric field

$$|\vec{E}|^2 = E_r^2 + E_\theta^2 = \left[\frac{V_0}{a}\left(\frac{r}{a} - 1\right)\right]^2 + \left[\frac{V_0}{r}\cos\theta\sin\theta\right]^2$$

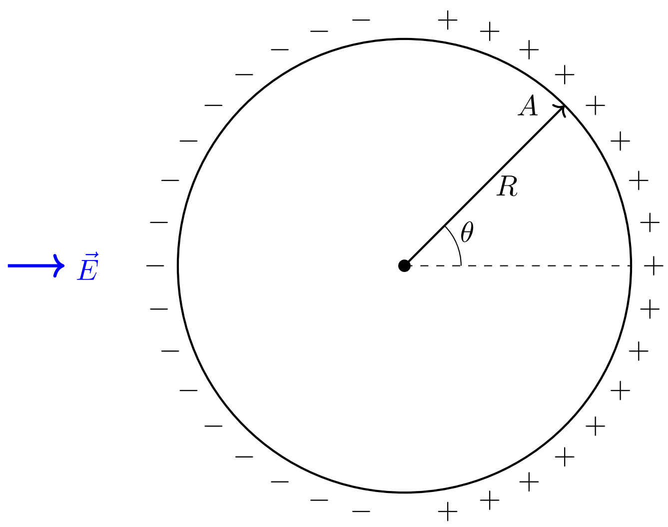
- For fixed θ, E_r grows linearly with r as $r \to \infty$, while $E_\theta \sim 1/r$ decays. - Therefore, at large distances:

$$|\vec{E}| \sim |E_r| \sim \frac{V_0}{a^2}r \quad \text{as } r \to \infty$$

- The electric field grows without bound as $r \to \infty$.

$$\boxed{\vec{E}(r,\theta) = \frac{V_0}{a}\left(\frac{r}{a} - 1\right)\hat{r} + \frac{V_0}{r}\cos\theta\sin\theta\,\hat{\theta}, \quad |\vec{E}| \text{ grows unbounded as } r \to \infty}$$

5. A solid conducting sphere of radius R is placed in a uniform electric field $\vec{E}$ as shown in Figure. Due to electric field non-uniform surface charges are induced on the surface of the sphere. Consider a point A on the surface of sphere at a polar angle θ from the direction of electric field as shown in Figure. Find the surface density of induced charges at point A in terms of electric field and polar angle θ.



Solution:

A neutral conducting sphere placed in a uniform external electric field $\vec{E}_0 = E_0\hat{z}$ develops induced surface charges. By symmetry, the induced charges are positive on one hemisphere and negative on the other, so the induced field has dipolar character.

The electric field outside the sphere can be written as the superposition of the applied field and the field of an induced dipole $\vec{p}$ at the center:

$$\vec{E} = \vec{E}_0 + \vec{E}_{\text{dip}}.$$

The electric field of a dipole is

$$\vec{E}_{\text{dip}} = \frac{1}{4\pi\varepsilon_0 r^3}\left(2p\cos\theta\,\hat{r} + p\sin\theta\,\hat{\theta}\right).$$

At the surface of a conductor, the tangential component of the electric field must vanish. The applied field has a tangential component

$$E_{0,\theta} = -E_0\sin\theta,$$

so at $r = R$,

$$\frac{p\sin\theta}{4\pi\varepsilon_0 R^3} = E_0\sin\theta.$$

This gives the induced dipole moment

$$p = 4\pi\varepsilon_0 R^3 E_0.$$

The radial component of the total electric field at the surface is then

$$E_r(R,\theta) = E_0\cos\theta + \frac{2p\cos\theta}{4\pi\varepsilon_0 R^3} = 3E_0\cos\theta.$$

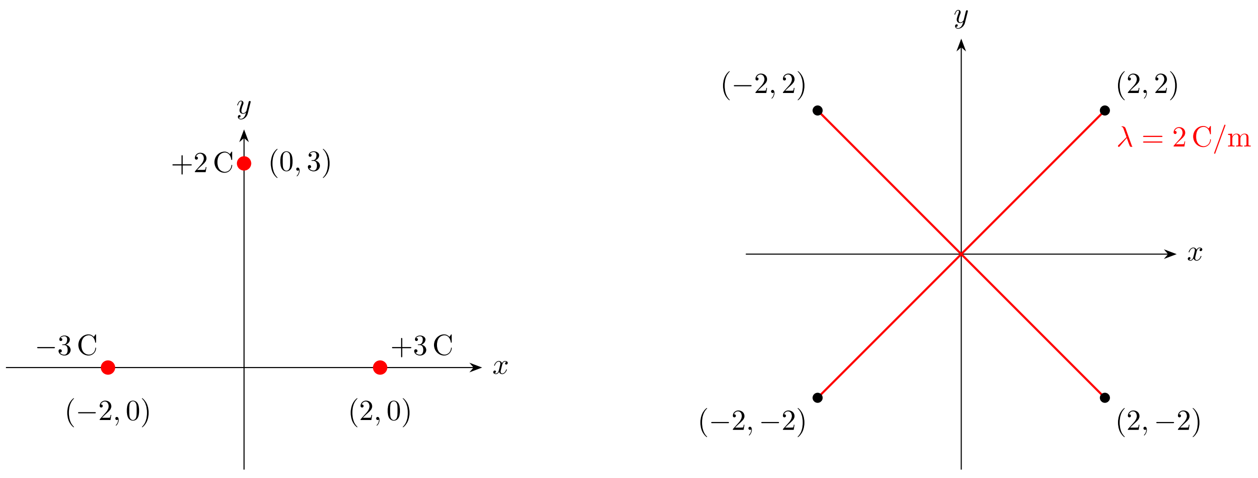
For a conductor, the surface charge density is related to the normal field by

$$\sigma(\theta) = \varepsilon_0 E_r(R,\theta).$$

Hence,

$$\boxed{\sigma(\theta) = 3\varepsilon_0 E_0\cos\theta}$$

6. Consider the below charge configurations:



(a) Discrete point charges (b) X-shaped line charge

(a) For these configurations, evaluate and plot the following quantities at the field points:
   i. Electric field $\vec{E}$
   ii. Electric potential V
   iii. Find out the energy required to assemble the configuration as in configuration (a).

(b) Verify through graphical visualization or representation:
   i Gauss' law
   ii $\vec{E} = -\nabla V$
   iii $\vec{E}$ is conservative

Solution:

(a) Let coulomb's constant is $k = \frac{1}{4\pi\varepsilon_0}$.

Configuration (a): Discrete Point Charges

Three point charges are located at

$$q_1 = -3 \text{ C at } (-2,0), \quad q_2 = +3 \text{ C at } (2,0), \quad q_3 = +2 \text{ C at } (0,3).$$

Electric Potential

$$V(x,y) = k\left[\frac{-3}{\sqrt{(x+2)^2+y^2}} + \frac{3}{\sqrt{(x-2)^2+y^2}} + \frac{2}{\sqrt{x^2+(y-3)^2}}\right].$$

Electric Field

$$\mathbf{E}(x,y) = k\left[-3\,\frac{(x+2)\hat{\mathbf{x}}+y\hat{\mathbf{y}}}{\left((x+2)^2+y^2\right)^{3/2}} + 3\,\frac{(x-2)\hat{\mathbf{x}}+y\hat{\mathbf{y}}}{\left((x-2)^2+y^2\right)^{3/2}} + 2\,\frac{x\hat{\mathbf{x}}+(y-3)\hat{\mathbf{y}}}{\left(x^2+(y-3)^2\right)^{3/2}}\right].$$

Energy of Assembly

$$U = k\sum_{i<j}\frac{q_iq_j}{r_{ij}}, \qquad r_{12}=4, \quad r_{13}=r_{23}=\sqrt{13}.$$

$$U = k\left(\frac{-9}{4} - \frac{6}{\sqrt{13}} + \frac{6}{\sqrt{13}}\right) = -\frac{9k}{4}.$$

Configuration (b): X-shaped Line Charge

Each arm carries a uniform line charge density

$$\lambda = 2 \text{ C/m}.$$

Electric Potential:

The electric potential at a point $\mathbf{r}$ due to a continuous charge distribution is

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{|\mathbf{r}-\mathbf{r}'|}$$

For a line charge with uniform linear charge density λ,

$$dq = \lambda\, dl'$$

and hence

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\lambda\, dl'}{|\mathbf{r} - \mathbf{r}'|}$$

Line charge 1: from $(-2, -2)$ to $(2, 2)$

The line lies along $y = x$ and is parameterized as

$$\mathbf{r}'(s) = (s, s), \qquad s \in [-2, 2]$$

The differential length element is

$$dl' = \sqrt{2}\, ds$$

Let the observation point be

$$\mathbf{r} = (x, y)$$

The distance between source and observation points is

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{(x - s)^2 + (y - s)^2}$$

The potential due to this line charge is

$$V_1(x, y) = \frac{\lambda\sqrt{2}}{4\pi\varepsilon_0} \int_{-2}^{2} \frac{ds}{\sqrt{(x - s)^2 + (y - s)^2}}$$

Integral simplification

The denominator can be rewritten as

$$(x - s)^2 + (y - s)^2 = 2\left[\left(s - \frac{x + y}{2}\right)^2 + \frac{(x - y)^2}{4}\right]$$

Hence,

$$V_1(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \int_{-2}^{2} \frac{ds}{\sqrt{\left(s - \frac{x+y}{2}\right)^2 + \frac{(x-y)^2}{4}}}$$

Using the standard result

$$\int \frac{du}{\sqrt{u^2 + a^2}} = \ln\left(u + \sqrt{u^2 + a^2}\right)$$

we obtain

$$V_1(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \left[\ln\left(s - \frac{x + y}{2} + \sqrt{\left(s - \frac{x + y}{2}\right)^2 + \frac{(x - y)^2}{4}}\right)\right]_{-2}^{2}$$

Line charge 2: from $(-2, 2)$ to $(2, -2)$

The line lies along $y = -x$ and is parameterized as

$$\mathbf{r}'(s) = (s, -s), \qquad s \in [-2, 2]$$

The differential length element is

$$dl' = \sqrt{2}\, ds$$

The distance between source and observation points is

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{(x - s)^2 + (y + s)^2}$$

The potential due to this line charge is

$$V_2(x, y) = \frac{\lambda\sqrt{2}}{4\pi\varepsilon_0} \int_{-2}^{2} \frac{ds}{\sqrt{(x - s)^2 + (y + s)^2}}$$

Integral simplification

The denominator becomes

$$(x - s)^2 + (y + s)^2 = 2\left[\left(s - \frac{x - y}{2}\right)^2 + \frac{(x + y)^2}{4}\right]$$

Hence,

$$V_2(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \int_{-2}^{2} \frac{ds}{\sqrt{\left(s - \frac{x-y}{2}\right)^2 + \frac{(x+y)^2}{4}}}$$

Evaluating the integral,

$$V_2(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \left[\ln\left(s - \frac{x - y}{2} + \sqrt{\left(s - \frac{x - y}{2}\right)^2 + \frac{(x + y)^2}{4}}\right)\right]_{-2}^{2}$$

By superposition, the total potential due to both line charges is

$$\boxed{V(x, y) = V_1(x, y) + V_2(x, y)}$$

Electric Field: The electric field due to a continuous line charge is given by Coulomb's law:

$$\vec{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\lambda\,(\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}\, dl'$$

where $\mathbf{r}$ is the observation point and $\mathbf{r}'$ is the source point.

Line charge 1: from $(-2, -2)$ to $(2, 2)$

This line lies along $y = x$. The source point is parameterized as

$$\mathbf{r}'(s) = (s, s), \qquad s \in [-2, 2]$$

The differential length element is

$$dl' = \sqrt{(ds)^2 + (ds)^2} = \sqrt{2}\, ds$$

Let the observation point be

$$\mathbf{r} = (x, y)$$

Then

$$\mathbf{r} - \mathbf{r}' = (x - s)\hat{x} + (y - s)\hat{y}$$
$$|\mathbf{r} - \mathbf{r}'| = \sqrt{(x - s)^2 + (y - s)^2}$$

Substituting into Coulomb's law,

$$\vec{E}_1(x, y) = \frac{\lambda\sqrt{2}}{4\pi\varepsilon_0} \int_{-2}^{2} \frac{(x - s)\hat{x} + (y - s)\hat{y}}{[(x - s)^2 + (y - s)^2]^{3/2}}\, ds$$

let $u = s - \frac{x+y}{2}$, then

Then

$$(x - s)^2 + (y - s)^2 = 2\left[u^2 + \frac{(x - y)^2}{4}\right]$$

Let's define unit vectors Along the line: $\hat{t} = \frac{1}{\sqrt{2}}(1, 1)$ and normal to the line: $\hat{n} = \frac{1}{\sqrt{2}}(1, -1)$ After integration, the tangential component vanishes and the field is purely normal.

Final electric field due to line 1

$$\vec{E}_1(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \frac{x - y}{|x - y|} \frac{1}{\sqrt{2}} \left[\frac{2 - \frac{x+y}{2}}{\sqrt{\left(2 - \frac{x+y}{2}\right)^2 + \frac{(x-y)^2}{4}}} + \frac{2 + \frac{x+y}{2}}{\sqrt{\left(2 + \frac{x+y}{2}\right)^2 + \frac{(x-y)^2}{4}}} \right] (\hat{x} - \hat{y})$$

Line charge 2: from $(-2, 2)$ to $(2, -2)$

This line lies along $y = -x$. The source point is parameterized as

$$\mathbf{r}'(s) = (s, -s), \qquad s \in [-2, 2]$$

The differential length element is

$$dl' = \sqrt{(ds)^2 + (-ds)^2} = \sqrt{2}\, ds$$

The displacement vector is

$$\mathbf{r} - \mathbf{r}' = (x - s)\hat{x} + (y + s)\hat{y}$$
$$|\mathbf{r} - \mathbf{r}'| = \sqrt{(x - s)^2 + (y + s)^2}$$

Substituting into Coulomb's law,

$$\vec{E}_2(x, y) = \frac{\lambda\sqrt{2}}{4\pi\varepsilon_0} \int_{-2}^{2} \frac{(x - s)\hat{x} + (y + s)\hat{y}}{[(x - s)^2 + (y + s)^2]^{3/2}}\, ds$$

Let $u = s - \frac{x-y}{2}$ then

$$(x - s)^2 + (y + s)^2 = 2\left[u^2 + \frac{(x + y)^2}{4}\right]$$

This mirrors the previous case with the replacement

$$x - y \longleftrightarrow x + y$$

Let's define unit vectors along the line: $\hat{t} = \frac{1}{\sqrt{2}}(1, -1)$. Normal to the line: $\hat{n} = \frac{1}{\sqrt{2}}(1, 1)$. After integration, the tangential component vanishes and the field is purely normal.

Final electric field due to line 2

$$\vec{E}_2(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \frac{x + y}{|x + y|} \frac{1}{\sqrt{2}} \left[\frac{2 - \frac{x-y}{2}}{\sqrt{\left(2 - \frac{x-y}{2}\right)^2 + \frac{(x+y)^2}{4}}} + \frac{2 + \frac{x-y}{2}}{\sqrt{\left(2 + \frac{x-y}{2}\right)^2 + \frac{(x+y)^2}{4}}}\right](\hat{x} + \hat{y})$$

By superposition,

$$\boxed{\vec{E}(x, y) = \vec{E}_1(x, y) + \vec{E}_2(x, y)}$$

Energy of the Line-Charge Configuration The electrostatic self-energy diverges due to the zero transverse dimension of the idealized line charge. A finite energy would be obtained only having discrete charge distribution.

(b)

i. Plot the divergence of electric field, we will observe that divergence is zero every where except the positions where charges are present.

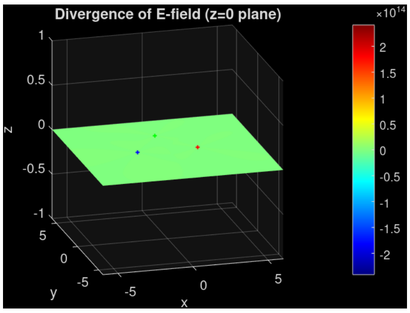


Figure 2: The divergence of Electric field is plotted for configuration-1 where except the charge locations it is zero and it is Dirac-delta distribution at charge locations

ii. Compare the plots of electric field from coulomb's law and using negative gradient of potential plot.

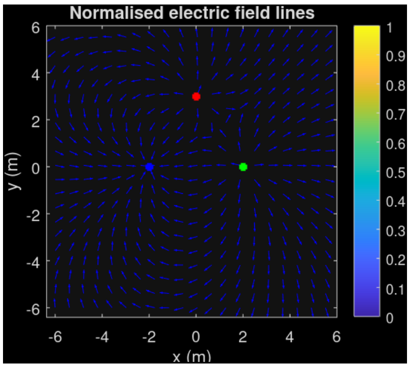


Figure 3: Electric field obtained from negative gradient of potential plot of configuration-(a)

This plot should match the electric field plot obtained in Part(a) of the question for configuration (a).

iii. In a conservative vector field, always $\nabla \times E = 0$. So we need to show that curl of electric field is zero. We directly use the curl function from matlab for plotting.

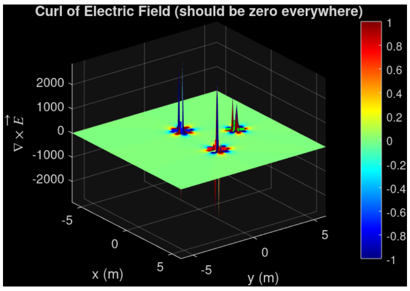


Figure 4: Negative Curl of Electric field is zero overall and the peaks near charges are only due to discretizations artifacts

7. A point charge of 12 nC is located at the origin. Four infinitely long uniform line charges are located parallel to the z-axis in the plane $x = 0$ as follows: Two line charges of density $80\,\mathrm{nC/m}$ at $y = -1m$ and $-5\,\mathrm{m}$, and two line charges of density $-50\,\mathrm{nC/m}$ at $y = -2\,\mathrm{m}$ and $-4\,\mathrm{m}$.

   (a) Determine the total electric flux crossing the plane $y = -3$, and state its direction.

   (b) Determine the total electric flux leaving the surface of a sphere of radius 4 m centered at $C(0, -3, 0)$.

   Solution:

   (a) Electric flux through the plane $y = -3$: The plane $y = -3$ divides space into two equal half-spaces. The point charge lies on one side. By symmetry, half of the total flux from the point charge crosses into the $-y$ half-space. Line charge contributions cancel due to symmetry. Thus, the flux crossing the plane is $Q_p/2 = 6$ nC directed in the $-\hat{\mathbf{a}}_y$ direction.

   (b) Electric flux leaving a sphere of radius 4 m// centered at $(0, -3, 0)$: By Gauss's law, $\Phi = Q_{\text{enclosed}}$. The sphere encloses the point charge $Q_p = 12$ nC. Translate the coordinate system so that the sphere is centered at the origin. The four infinite line charges are now located at

$$y = \pm 1, \quad y = \pm 2.$$

   Only the portions of the line charges lying inside the sphere contribute to the flux.

   For a line charge at perpendicular distance d from the center, the length inside the sphere is

$$h = 2r \sin\left(\cos^{-1}\frac{d}{r}\right).$$

   Line charges at $d = 1$ m ($\lambda = -50$ nC/m)

$$h_1 = 2(4)\sin\left(\cos^{-1}\frac{1}{4}\right) = 7.746 \text{ m}.$$

$$Q_1 = 2(7.746)(-50) = -774.6 \text{ nC}.$$

Line charges at $d = 2$ m ($\lambda = 80$ nC/m)

$$h_2 = 2(4)\sin\left(\cos^{-1}\frac{2}{4}\right) = 6.928 \text{ m}.$$

$$Q_2 = 2(6.928)(80) = 1108.5 \text{ nC}.$$

Total enclosed charge

$$Q_{\text{enc}} = Q_p + Q_1 + Q_2 = 12 - 774.6 + 1108.5 = 345.9 \text{ nC}.$$

Hence, the outward electric flux is

$$\boxed{\Phi = 346 \text{ nC}}$$